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MODCAN: driver gene identification based on multi-omics features and differential co-association networks for tumor subtypes

Ponian Li[1], Guodong Xiao[1], Haihui Wang[1], Chunrui Xu[2] and Yusen Zhang[1*]

*Correspondence:
Yusen Zhang
zhangys@sdu.edu.cn
[1]School of Mathematics and Statistics, Shandong University, Weihai 264209, China
[2]School of Life Sciences, Zhengzhou University, Zhengzhou 450001, China

Abstract

Background Despite the identification of some pan-cancer driver genes through international collaborative initiatives, the discovery of key cancer driver genes remains a formidable challenge. This limitation continues to hinder progress in critical areas, such as early diagnosis, prognostic evaluation, and precision medicine. Consequently, the accurate identification of key driver genes for specific cancer types has become a central focus of bioinformatics research.

Results We present MODCAN, a novel semi-supervised algorithm based on multi-omics features and differential co-association networks. MODCAN facilitates the meaningful stratification of tumor samples into distinct subtypes, enabling a comprehensive exploration of multi-omics features that reflect the inherent heterogeneity of tumors. By constructing differential co-association networks, MODCAN reveals the unique genetic interactions characteristic of each subtype, thereby facilitating the complementary integration of information.

Conclusions When applied to ten cancer datasets from TCGA, MODCAN significantly outperforms both existing supervised and unsupervised learning algorithms, exhibiting superior performance in terms of precision, recall, and AUPR. Furthermore, MODCAN demonstrates substantial advantages in predicting potential tumor-specific driver genes. Notably, these genes not only exhibit strong specificity for their respective cancers, but also reveal tumor heterogeneity across distinct subtypes.

Keywords Tumor-specific driver gene, Multi-omics feature extraction, Differential co-association network, Multi-graph convolutional network

Background

Cancer arises from a series of genetic and transcriptomic alterations in normal cells, leading to aberrant differentiation, dysregulated growth, invasion, and metastasis—distinctive biological features that disrupt the equilibrium between cell proliferation and apoptosis [1, 2]. Genomic mutations encompass both single nucleotide variants (SNVs) and chromosomal aberrations. Specific mutations confer a selective growth advantage to cells, fostering the uncontrolled proliferation of malignant clones [3]. These pivotal

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mutations are referred to as driver mutations, and their identification is critical for early cancer diagnosis, targeted therapies, and precision medicine. Research has demonstrated that alterations in gene expression are often driven by somatic mutations in the cancer genome [4], with distinct transcriptomic signatures in different cancer types closely associated with specific DNA mutation patterns [5]. Thus, integrating genomic data through the lens of tumor heterogeneity, and extracting meaningful insights from these complex datasets, emerges as a promising method for identifying cancer driver genes.

In the early stages of cancer research, researchers have sought to identify potential driver genes by detecting those with mutation rates significantly elevated or reduced relative to the background mutation rate. For example, MuSiC [6] integrates gene expression data with mutation frequencies, enabling the precise identification of low-frequency driver mutations that might otherwise be overlooked. Similarly, MutSigCV [7] analyzes exome sequences from tumor-normal sample pairs to uncover significant alterations in mutation frequency and spectra. Meanwhile, DriverPower [8] employs functional impact scores alongside mutation burden to identify driver mutations in both coding and non-coding regions of the cancer genome. However, mutation rate-based algorithms for identifying cancer driver genes may be prone to increased false positives due to the high prevalence of passenger mutations.

To address the challenges outlined above, researchers have shifted their focus toward biological networks of gene interactions, such as the protein-protein interaction (PPI) network, transcriptional regulatory network, and differential co-expression network. For example, MaxMIF [9] identifies key cancer driver genes by maximizing the mutual information flow between genes while integrating the PPI network. DriverGroup [10] utilizes the controllability theory of complex networks to pinpoint minimal control sets within the transcriptional regulatory network. Meanwhile, WMDS.net [11] constructs the weighted differential co-expression network to identify key driver gene nodes in cancer. Moreover, several network-based algorithms not only examine the interactions of the genome within biological networks but also explore the mutational landscape of the genome. For instance, CovEx [12] identifies cancer driver gene modules by analyzing the coverage and exclusivity of gene mutations, employing binary linear programming. Similarly, MaxCLK [13] identifies key driver gene modules in cancer through the analysis of k-cliques and information entropy within the exclusivity graph.

In recent years, with the rise of machine learning and neural networks, network-based algorithms have seen significant progress. For example, DORGE [14] integrates genetic and epigenetic features and employs a logistic regression classifier to identify oncogenes and tumor suppressor genes. DriverRWH [15], by constructing the hypergraph network and integrating mutation data with the PPI network, employs the hypergraph random walk algorithm to identify cancer driver genes and rank their importance. While these algorithms effectively reduce the impact of passenger mutations, they still face challenges in fully extracting reliable information from existing genomic data. Additionally, MTGCN [16] incorporates gene features into the PPI network and optimizes both node prediction and link prediction tasks within a multi-task learning framework to accurately identify cancer driver genes. Furthermore, GGraphSAGE [17] combines graph attention networks with GraphSAGE layers, integrating genomic data and network-derived features to identify key driver genes in cancer.

Despite the identification of some pan-cancer driver genes through international collaborative initiatives, the discovery of key cancer driver genes remains a formidable challenge. This limitation continues to hinder progress in critical areas, such as early diagnosis, prognostic evaluation, and precision medicine. Consequently, the accurate identification of key driver genes for specific cancer types has become a central focus of bioinformatics research. On the one hand, the effective extraction of gene features from genomic data remains a crucial prerequisite for network-based driver gene prediction. On the other hand, the transition of cells from a normal to a diseased state is often accompanied by significant reorganization of transcriptional co-association networks. Therefore, we present MODCAN, a novel semi-supervised algorithm based on multi-omics features and differential co-association networks. MODCAN facilitates the meaningful stratification of tumor samples into distinct subtypes, enabling a comprehensive exploration of multi-omics features that reflect the inherent heterogeneity of tumors. By constructing differential co-association networks, MODCAN reveals the unique genetic interactions characteristic of each subtype, thereby facilitating the complementary integration of information.

When applied to ten cancer datasets from TCGA [18], MODCAN not only successfully identifies driver genes but also uncovers their differential significance across specific cancer types. Remarkably, MODCAN significantly outperforms both existing supervised and unsupervised learning algorithms, exhibiting superior performance in terms of precision, recall, and AUPR. Furthermore, MODCAN demonstrates substantial advantages in predicting potential tumor-specific driver genes. For example, RYR2 exhibits significant mutation differences in the immune subtype of BLCA. Additionally, PI3K inhibition enhances anticancer effects via cytotoxic autophagy in BRCA. The analysis illustrates that these driver genes not only exhibit strong specificity for their respective cancers but also reveal tumor heterogeneity across distinct subtypes, highlighting their potential as novel biomarkers and therapeutic targets.

Methods

Overview

MODCAN is a semi-supervised algorithm based on multi-omics features and differential co-association networks, designed to identify driver genes for specific cancer types, as illustrated in Fig. 1. MODCAN initially constructs the individualized gene network through co-expression communities, yielding community cohesion scores for each tumor sample. Subsequently, tumor samples are stratified into distinct subtypes using genomic data, with global and local features generated for mutated genes. To model shifts in gene correlations within tissue samples during carcinogenesis, differential co-association networks are derived from the PPI network. Finally, multi-graph convolutional networks are applied to aggregate gene features and facilitate the identification of cancer driver genes at the cohort level.

Data collection and preprocessing

The genomic data employed by MODCAN include gene expression, promoter methylation, copy number variation (CNV), and single nucleotide variation (SNV) data. These datasets were derived from TCGA cancer patient data, which were accessed via the UCSC Xena Browser [19], spanning ten distinct cancer types, as outlined in Table 1.

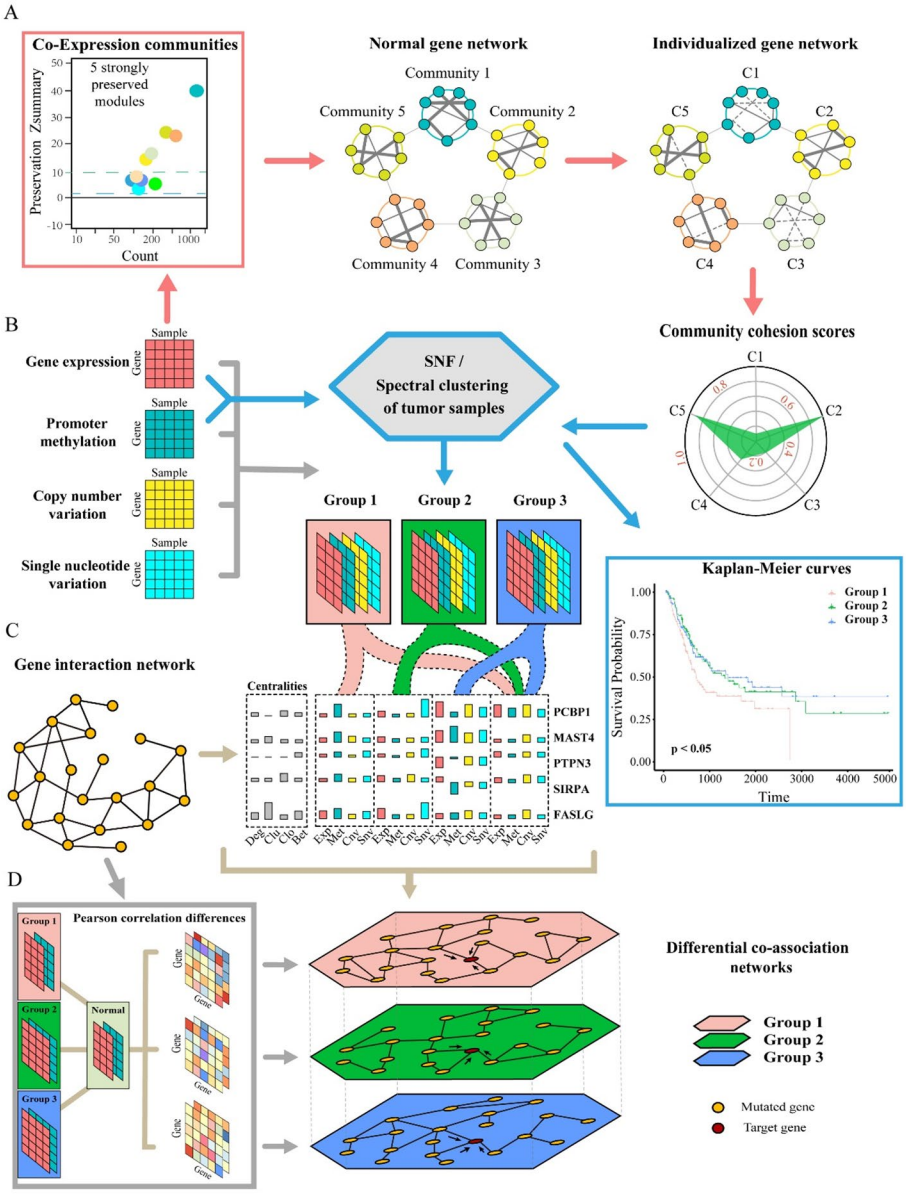

Fig. 1 Workflow of the MODCAN model. **A** Community cohesion scores of tumor samples. MODCAN retains only those co-expression communities with $Zsummary > 10$, along with the top 10% of interactions by weight within each community. **B** Subtype clustering of tumor samples. Tumor samples corresponding to distinct subtypes exhibit significantly different survival probabilities in Kaplan-Meier curves. **C** Multi-omics feature extraction of mutated genes. MODCAN extracts local features from genomic data specific to the corresponding tumor subtypes, while concurrently deriving global features from comprehensive genomic data and the PPI network. **D** Construction of differential co-association networks and multi-graph convolutional networks. The multiplex network shares the same gene nodes as the PPI network, but for any pair of genes, multiple interactions exist, effectively integrating differential gene correlations across different tumor subtypes into a unified graph

RNA-seq expression data obtained from the Illumina platform were log-transformed ($\log_2(\text{count} + 1)$), while methylation data were sourced from the TCGA 450k Illumina BeadChip arrays. The analysis focused on cancer types in TCGA that include both tumor and normal samples, with each of the ten cancer types containing at least 20 normal samples. For each cancer type, genes without mutations in tumor samples and those with missing genomic data across all tumor samples were excluded. Tumor and normal samples with publicly available genomic data were retained for analysis.

Table 1 Mutated genes and samples in cancer datasets

Cancer	Mutated genes	Tumor samples	Normal samples	Cancer	Mutated genes	Tumor samples	Normal samples
BLCA	18,551	411	21	KIRP	17,537	274	45
BRCA	17,465	780	96	LUAD	18,057	463	32
COAD	18,431	308	38	LUSC	18,541	369	42
HNSC	18,207	502	50	PRAD	18,254	497	50
KIRC	17,490	320	160	UCEC	18,647	431	46

Table 2 Mutated genes and communities in cancer datasets

Cancer	Mutated genes	Communities	Cancer	Mutated genes	Communities
BLCA	13,264	7	KIRP	13,200	12
BRCA	13,544	24	LUAD	16,000	24
COAD	10,751	36	LUSC	13,691	32
HNSC	9034	27	PRAD	10,741	6
KIRC	14,333	24	UCEC	14,132	31

Furthermore, we employed the latest PPI network Stringv12, sourced from the STRING-db [20]. To ensure high confidence in gene associations, only the top 70% of gene interactions, ranked by confidence scores, were retained. Despite this filtering, the simplified Stringv12 network still encompasses 18,583 genes and 4,140,992 gene interactions, sufficiently capturing the majority of mutated genes across the analyzed cancer types.

Community cohesion scores of tumor samples

The community cohesion score [21] provides a precise measure of functional degradation within each community of the individualized gene network. In the absence of prior knowledge for further tumor subtype stratification, this score serves as a valuable tool for the subsequent rational classification of tumor samples into distinct subtypes.

Normal gene network and co-expression communities

The normal samples are divided into a training set (80%) and a test set (20%), with parameters (pickSoftThreshold) determined on the training set. An adjacency matrix (adjacency) and a weight matrix (TOMsimilarity) are then computed to construct the normal gene network. The weight matrix is subsequently used as input for hierarchical clustering (hclust, cutreeDynamic) to identify highly correlated co-expression communities. Additionally, permutation testing is performed on the test set to retain only those co-expression communities with $Zsummary > 10$, along with the top 10% of interactions by weight within each community. Mutated genes within these communities are then selected from the cancer datasets (see Table 2). Further details of the process can be found in Fig. S1.

Perturbation interactions and individualized gene network

MODCAN employs the improved median rank score (MRS) to normalize gene expression data, effectively mitigating batch effects (see Fig. S2). Within the normal gene network, each gene interaction is modeled using linear regression ($\widehat{Y} = \beta_0 + \beta_1 X$) to simulate the correlation between two genes. The difference

between the observed expression values and the predictions from the linear regression model is then used to calculate the vertical distance distribution for normal samples.

For each tumor sample, the vertical distance $d_j = \frac{|\beta_1 X_j - Y_j + \beta_0|}{\sqrt{\beta_1^2 + 1}}$ is computed and subsequently transformed into the $Modified\, zscore_j = \frac{d_j - \tilde{d}}{1.486 * MAD}$, based on the median $\tilde{d}$ and the MAD of the vertical distance distribution. Interactions with the one-tailed P-value below 0.001 are considered individualized perturbation interactions. These perturbation interactions are then removed to construct the individualized gene network.

Network efficiency and community cohesion scores

MODCAN utilizes network efficiency to assess the impact of perturbation interactions on the cohesion within communities of the individualized gene network. The network efficiency for the k-th community in each individualized gene network is calculated as follows:

$$\text{Network efficiency}_k = \frac{1}{N(N-1)} \sum_{X \neq Y \in G_k} \frac{1}{d(X,Y)} \tag{1}$$

Here, G_k and N denote the set of nodes and the total number of nodes within the k-th community, respectively. $d(X, Y)$ represents the length of the weighted shortest path between nodes X and Y. If two nodes are disconnected within the network, $d(X, Y)$ is assigned the value of 0.

Finally, the community cohesion score for the k-th community is computed using Eq. (2). Each tumor sample is then assigned the community cohesion scores corresponding to the number of co-expression communities.

$$\text{Community cohesion score}_k = \frac{\text{Network efficiency}_k \text{ in the individualized gene network}}{\text{Network efficiency}_k \text{ in the normal gene network}} \tag{2}$$

Subtype clustering of tumor samples

To extract effective gene features associated with tumor subtypes from genomic data, MODCAN first selects the top 30 most variable genes from both gene expression and methylation data for each cancer type. Based on these selected genes and community cohesion scores, the corresponding Euclidean distance matrices are calculated as inputs for the SNF algorithm, and then transformed into similarity matrices for sample pairs, which are integrated into an affinity matrix M for tumor samples. In this matrix, $M_{ij} \in [0, 1]$ represents the final similarity between tumor samples i and j. Further details regarding the SNF algorithm can be found in Fig. S3.

Finally, MODCAN applies spectral clustering to the affinity matrix M, generating clustering results for tumor samples. Tumor samples within the same cluster are classified as belonging to the same subtype. Based on these subtypes, genomic data are stratified into distinct categories, enabling the extraction of gene features specific to each subtype, as shown in Table 3. Tumor samples corresponding to distinct subtypes exhibit significantly different survival probabilities in Kaplan-Meier curves (see Fig. S4).

Multi-omics feature extraction of mutated genes

The effective extraction of gene features from genomic data remains a crucial prerequisite for network-based driver gene prediction [22, 23]. MODCAN facilitates cancer

Table 3 Tumor subtypes and features in cancer datasets

Cancer	Tumor subtypes	Features	Cancer	Tumor subtypes	Features
BLCA	3	20	KIRP	6	32
BRCA	5	28	LUAD	3	20
COAD	4	24	LUSC	6	32
HNSC	5	28	PRAD	3	20
KIRC	5	28	UCEC	5	28

driver gene identification by focusing on features that comprehensively capture tumor heterogeneity, rather than relying on pan-cancer datasets for feature generation.

Given a specific cancer type, MODCAN assigns L subtypes to the tumor samples via spectral clustering, with each subtype comprising n_l tumor samples. Let $E^{(l)}$ and $M^{(l)}$ represent the gene expression and methylation matrices for tumor subtype $l(l = 1, 2, \ldots, L)$, where the rows correspond to mutated genes and the columns correspond to tumor samples. Similarly, $S^{(l)}$ and $C^{(l)}$ denote the binary SNV and CNV matrices for tumor subtype l, respectively. For each tumor subtype l, all matrices share the same row and column labels. Notably, let n_0 represents the total number of tumor samples, with the four matrices covering all tumor samples being denoted as $E^{(0)}$, $M^{(0)}$, $S^{(0)}$, and $C^{(0)}$.

Extraction from the genomic data of the corresponding tumor subtype

The difference in average expression and methylation values between tumor and normal samples for gene i in subtype l is calculated as follows:

$$FE_i^{(l)} = \frac{1}{n_l} \sum_j^{n_l} E^{(l)}(i,j) - e_{nor}(i) \tag{3}$$

$$FM_i^{(l)} = \frac{1}{n_l} \sum_j^{n_l} M^{(l)}(i,j) - m_{nor}(i) \tag{4}$$

Here, $e_{nor}(i)$ and $m_{nor}(i)$ denote the average expression and methylation values of gene i in normal samples, respectively.

The SNV frequency $FS_i^{(l)}$ and CNV frequency $FC_i^{(l)}$ of gene i in subtype l are defined as follows:

$$FS_i^{(l)} = \frac{1}{n_l} \sum_{j=1}^{n_l} S^{(l)}(i,j) \tag{5}$$

$$FC_i^{(l)} = \frac{1}{n_l} \sum_{j=1}^{n_l} C^{(l)}(i,j) \tag{6}$$

Here, $S^{(l)}(i,j) = 1$ indicates that gene i is mutated in sample j, and $C^{(l)}(i, j) = 1$ indicates the presence of copy number alteration for gene i in sample j.

The gene features generated by MODCAN from the genomic data of the corresponding tumor subtypes are referred to as local features, whereas those derived from the complete genomic data (i.e., $l = 0$) contribute to the global features.

Calculation of network features from the PPI network

It is widely recognized that driver genes tend to exhibit higher node degrees within the PPI network. Furthermore, local network density reflects the intricate interactions between mutated genes and their neighbors. Therefore, MODCAN employs various centrality metrics within the PPI network as network features, including

clustering coefficient, closeness centrality, degree centrality, and betweenness centrality. Fig. S5 illustrates the disparities in centrality metrics between driver and non-driver genes, underscoring their unique roles in network positioning and interactions.We designate the gene features derived from the complete genomic data and the network features calculated from the PPI network as global features. MODCAN concatenates both the global and local feature sets into a single feature matrix $X \in \mathcal{R}^{m \times d}$, where m represents the number of mutated genes, and the feature dimension $d = 4\,(L+2)$, proportional to the number of tumor subtypes. To standardize the value ranges across different features, MODCAN normalizes the feature matrix X using Min-Max scaling.

Construction of differential co-association networks

The transition of cells from a normal to a diseased state is often accompanied by significant reorganization of transcriptional co-association networks [24]. To capture the transcriptional differences across distinct tumor subtypes, MODCAN constructs differential co-association networks by selectively removing specific protein interactions from the PPI network and subsequently reweighting the network.

Given a specific cancer type, let $E^{(l)}$ and $M^{(l)}$ represent the gene expression and methylation data for tumor samples in subtype l, respectively, while E^{nor} and M^{nor} correspond to the same data from normal samples. MODCAN computes the Pearson correlation matrices between genes for tumor and normal samples, denoted as $P_E^{(l)}$, $P_M^{(l)}$, P_E^{nor}, and P_M^{nor}. The correlation difference matrices are defined as $\Delta P_E^{(l)} = |P_E^{(l)} - P_E^{nor}|$ and $\Delta P_M^{(l)} = |P_M^{(l)} - P_M^{nor}|$, reflecting the magnitude of gene correlation changes between normal and cancerous states for tumor subtype l. MODCAN sets a threshold θ to assess the significance of these correlation changes. For values exceeding θ in $\Delta P_E^{(l)}$ and $\Delta P_M^{(l)}$, the corresponding gene correlation changes are of greater research significance, while those below θ are set to zero.

The interaction weights within the PPI network A are then redistributed as follows:

$$A'(i,j) = \begin{cases} \text{sigmoid}\left(\frac{A(i,j)}{A_{max}}\right), & \text{if } A(i,j) > 0 \\ 0, & \text{otherwise} \end{cases} \tag{7}$$

The interaction weight between genes g_i and g_j is denoted as $A(i,j)$ (where $A(i,j) \geq 0$), with A_{max} representing the maximum weight.

Finally, MODCAN combines the correlation difference matrices $\Delta P_E^{(l)}$ and $\Delta P_M^{(l)}$ with the PPI network A' (where $\omega_1 + \omega_2 = 1$), thereby constructing the differential co-association network for tumor subtype l using Eq. (8).

$$A^{(l)} = A'\left(\omega_1 \Delta P_E^{(l)} + \omega_2 \Delta P_M^{(l)}\right) \tag{8}$$

The differential co-association network $A^{(l)}$ for tumor subtype l retains only those gene interactions present in the PPI network that exhibit significant changes in gene correlations between normal and cancerous states. Notably, the settings of ω_1 and ω_2 significantly influence MODCAN's performance in identifying cancer driver genes.

Construction of multi-graph convolutional networks

MODCAN introduces the multi-graph convolutional networks (MGCN). The weighted multiplex network is constructed by concatenating the differential co-association networks $A^{(l)}$ (for $l = 1, 2, \ldots, L$), represented in tensor form as $A_{multi} \in \mathcal{R}^{m \times m \times L}$. The multiplex network A_{multi} shares the same gene nodes as the PPI network, but for any pair of genes, multiple interactions exist, effectively integrating differential gene correlations across different tumor subtypes into a unified graph.

The multi-graph Laplacian matrix L_{multi}, whose slices $L_{multi}^{(l)}$ are derived from the corresponding differential co-association networks $A^{(l)}$, satisfies the following condition:

$$L_{\text{multi}}^{(l)} = \tilde{D}_l^{-1/2} \tilde{A}^{(l)} \tilde{D}_l^{-1/2} \tag{9}$$

Here, $\tilde{A}^{(l)} = A^{(l)} + I$ denotes the differential co-association network with self-loops added, while $\tilde{D}_l$ is a diagonal matrix, where $\tilde{D}_l(i,i) = \sum_j A^{(l)}(i,j)$.

Thus, the multi-graph convolution rule in MODCAN is defined as follows:

$$H^{(r+1)} = \sigma\left(L_{multi} \otimes H^{(r)} W^{(r)}\right) \tag{10}$$

$$L_{multi} \otimes H^{(r)} = concatenate\left\{L_{multi}^{(l)} H^{(r)} | l = 1, 2, \ldots, L\right\} \tag{11}$$

Here, $H^{(0)} = X$ denotes the input feature matrix, W denotes the learnable weight matrix, with the activation function $\sigma(\cdot)$ defined as ReLU. Notably, $concatenate\{\cdot\}$ refers to the concatenation of the input matrix sequence into a single matrix. Subsequently, the MGCN facilitates the exchange of information across these differential co-association networks through the weight matrix W.

MODCAN performs cancer driver gene identification at the cohort level using two multi-graph convolutional layers, followed by a single hidden layer feedforward neural network. The resulting output is an $m \times 2$ matrix, transformed via the $SoftMax$ function, which provides the probabilities of each mutated gene being classified as either a driver or a non-driver gene.

Curation of positive and negative genes

We obtained 579 known cancer driver genes from the COSMIC Cancer Gene Census (CGC v99) [25] as positive genes for training MODCAN. Subsequently, we iteratively excluded genes implicated in cancer initiation and progression, selecting 2,179 non-driver genes from the NCG [26], COSMIC, and OMIM databases, as well as KEGG cancer pathways.

Model training

In training the MODCAN model, labeled data are randomly split into training (75%) and testing (25%) sets using stratified sampling, ensuring consistent proportions of positive and negative genes in both sets. The training set is further divided into 10 subsets, with a 9:1 ratio for model training and validation, enabling 10-fold cross-validation.

For model training, the multi-graph Laplacian matrix $L_{multi} \in \mathcal{R}^{m \times m \times L}$, the feature matrix of mutated genes $X \in \mathcal{R}^{m \times d}$, and the gene label vector $y \in \mathcal{R}^{m}$ are utilized. The cross-entropy loss for the training nodes is defined as follows:

$$L = \sum_{g_t} -\alpha(y_t \log(\tilde{y}_t) + (1 - y_t) \log(1 - \tilde{y}_t)) \tag{12}$$

Here, g_t refers to the labeled gene used in model training, while $\tilde{y}_t$ represents the predicted probability that g_t is classified as a driver gene. The true label of g_t, denoted as $y_t \in \{0, 1\}$, indicates whether g_t is a driver gene when $y_t = 1$, or a non-driver gene when $y_t = 0$.

Given the class imbalance between positive and negative genes in the training set, MODCAN introduces a parameter α, which amplifies the loss for positive genes by a factor of two, while leaving the loss for negative genes unchanged. Additionally, the weight matrix W in MODCAN is updated using the ADAM optimizer [27], with the maximum number of training epochs set to 1000. The model selects the parameter set that yields the best performance on the validation set from all local optima, which is then used as the final trained model.

Results

MODCAN identifies driver genes accurately

To assess the advantages of constructing differential co-association networks, we designed GCN [28] and GAT [29] models, which were trained using the same feature matrix as MODCAN, ensuring consistency in hyperparameter settings. Additionally, we compared MODCAN with the latest supervised learning algorithms, including MTGCN and GGraphSAGE, to demonstrate its superiority in extracting multi-omics features from mutated genes. For each cancer type, all supervised learning algorithms were trained and evaluated using the same training and test sets.

Furthermore, we compared MODCAN with two advanced, network-based unsupervised learning algorithms, DriverRWH and MaxMIF. To facilitate a performance comparison, we extracted the predicted scores or rankings for mutated genes from the outputs of these algorithms. It is important to note that all network-based algorithms utilized the PPI network Stringv12, ensuring that the comparison of cancer driver gene identification performance was not influenced by discrepancies between networks.

The AUPR values of the various algorithms on the same test set are depicted in Fig. 2. Across ten cancer types, MODCAN consistently achieved the highest AUPR values. Notably, under the same feature matrix, MODCAN consistently outperforms both the GCN and GAT models. This underscores the effectiveness of differential co-association networks based on distinct tumor subtypes, which complement multi-omics features and substantially enhance MODCAN's performance in identifying cancer driver genes.

As shown in Fig. 2, MODCAN consistently outperforms the latest MTGCN and GGraphSAGE algorithms. This finding underscores that, rather than merely incorporating additional gene features, a more thorough exploration of genomic data, which more comprehensively captures tumor heterogeneity, can more effectively enhance MODCAN's performance in identifying cancer driver genes. Moreover, supervised driver gene identification algorithms typically achieve higher precision than their unsupervised

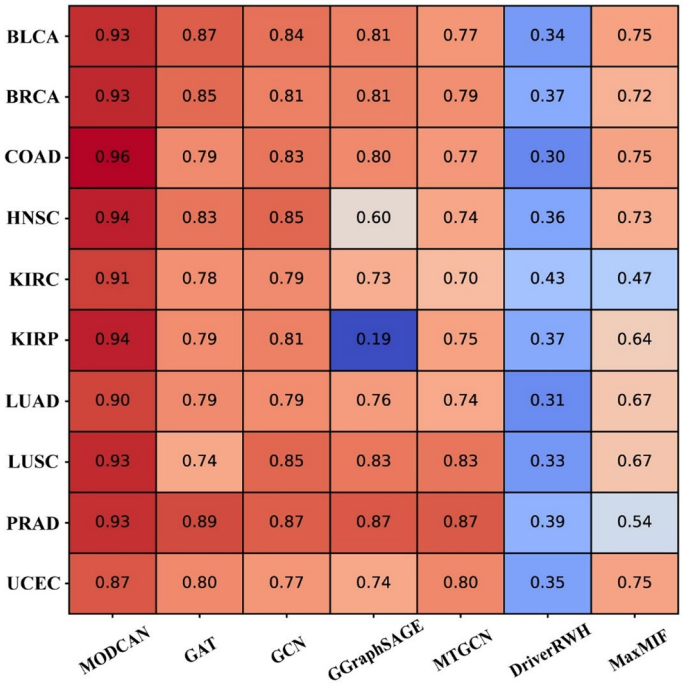

Fig. 2 Comparison of AUPR values for various algorithms across ten cancer types

counterparts. In summary, MODCAN demonstrates outstanding performance in cancer driver gene identification, surpassing existing advanced algorithms.

To evaluate the performance of various algorithms in identifying cancer driver genes across the complete datasets, we utilized the gold standard set of 579 known driver genes from CGCv99. ROC curves were generated to represent the complete predictive outcomes of each algorithm for different cancer types. As shown in Fig. 3, MODCAN consistently achieved the highest AUC values across all cancer types, with AUCs exceeding 0.90 in seven of them.

The AUC values for each algorithm across ten cancer types are provided in Table 4. The exceptional performance of MODCAN on both the test and complete datasets highlights its superiority in cancer driver gene identification. This success can be attributed to the algorithm's comprehensive exploration of multi-omics features that reflect tumor heterogeneity, as well as the construction of differential co-association networks based on distinct tumor subtypes, facilitating the complementary integration of information.

Ablation experiments of MODCAN

To assess the contribution of multi-omics features and differential co-association networks to the algorithm's performance in cancer driver gene identification, we conducted ablation experiments on MODCAN. For ten cancer types, we progressively removed specific subsets of multi-omics features or parts of differential co-association networks, quantifying the differences in performance on the test set. As shown in Fig. 4, the removal of any component led to varying degrees of reduction in AUPR values. These results further highlight the indispensable role of MODCAN in accurately identifying cancer driver genes.

In MODCAN, removing differential co-association networks derived from either expression or methylation data resulted in a significant decrease in AUPR values. These

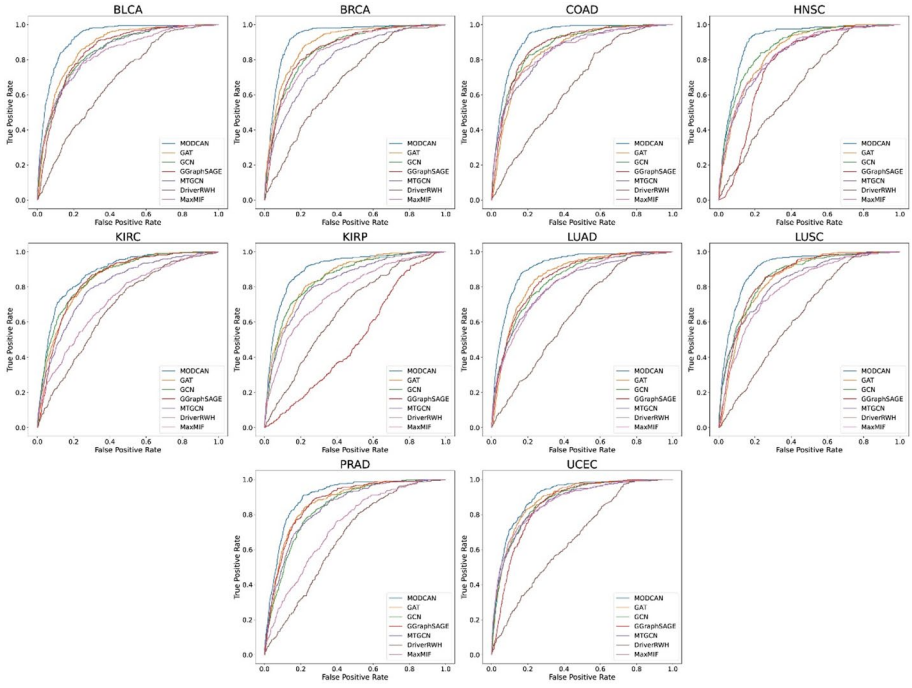

Fig. 3 ROC curves for various algorithms across ten cancer types

Table 4 Comparison of AUC values for various algorithms across ten cancer types

Cancer	MODCAN	GAT	GCN	GGraphSAGE	MTGCN	DriverRWH	MaxMIF
BLCA	0.9261	0.8780	0.8550	0.8578	0.8475	0.6846	0.8358
BRCA	0.9316	0.8925	0.8620	0.8632	0.8029	0.6872	0.8496
COAD	0.9272	0.8567	0.8779	0.8785	0.8524	0.6619	0.8506
HNSC	0.9141	0.8554	0.8764	0.7804	0.8277	0.6592	0.8346
KIRC	0.8809	0.8477	0.8584	0.8426	0.8102	0.6935	0.7358
KIRP	0.9166	0.8590	0.8534	0.5062	0.8328	0.6728	0.7714
LUAD	0.9145	0.8601	0.8384	0.8514	0.8146	0.6522	0.8142
LUSC	0.9112	0.8447	0.8600	0.8559	0.8325	0.6461	0.7986
PRAD	0.8980	0.8638	0.8378	0.8664	0.8388	0.6693	0.7305
UCEC	0.8984	0.8855	0.8742	0.8551	0.8669	0.6620	0.8707

findings emphasize that differential co-association networks effectively reveal the unique genetic interactions characteristic of each subtype, providing 'latent variables' for driver gene identification. Moreover, removing gene features derived from genomic data related to tumor subtypes caused a further decline in AUPR values, highlighting the importance of features that comprehensively capture tumor heterogeneity to enhance MODCAN's performance in identifying driver genes. Notably, removing network features calculated from the PPI network reduced MODCAN's stability, adversely affecting the precision of driver gene identification.

Prediction of candidate driver genes

To validate that the driver genes predicted by MODCAN are strongly associated with their respective cancers and play key roles in cancer initiation and progression, we defined mutated genes with a predicted probability ≥ 0.95 as candidate driver genes. These candidate genes were then compared with known driver genes from the CGC and

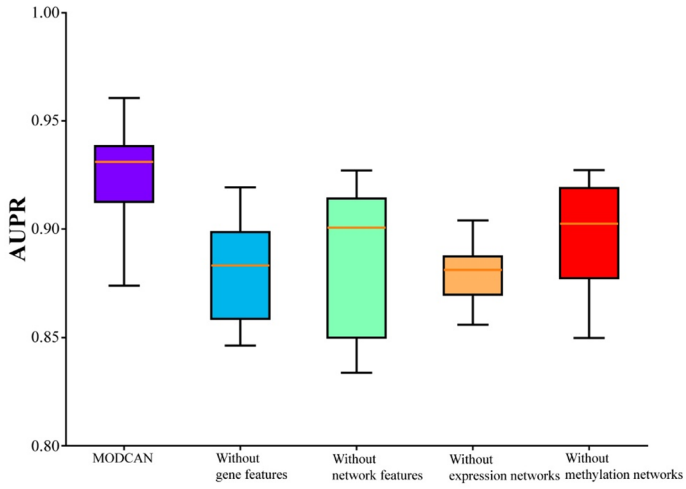

Fig. 4 AUPR values on the test set from MODCAN ablation experiments. The horizontal axis illustrates the full MODCAN algorithm, along with the results obtained after removing gene features derived from genomic data related to tumor subtypes, network features calculated from the PPI network, and differential co-association networks derived from either expression or methylation data

Table 5 Number of candidate and known driver genes in various cancer types

Cancer	Number of candidate genes	Number of candidate genes in CGC	Number of candidate genes in IntOGen
BLCA	419	139	44
BRCA	518	137	42
COAD	423	121	11
HNSC	550	145	21
KIRC	415	102	8
KIRP	473	135	10
LUAD	556	158	30
LUSC	377	122	16
PRAD	473	120	31
UCEC	431	126	20

IntOGen [30] databases. The number of candidate and known driver genes for each cancer type is shown in Table 5. Notably, many known tumor-specific driver genes were identified among the candidate genes predicted by MODCAN. For example, of the 93 known tumor-specific driver genes in BLCA, 44 were successfully identified, while in LUAD, 30 out of 77 were predicted as candidate driver genes by MODCAN.

We performed KEGG cancer pathway enrichment analysis for the candidate driver genes in each cancer type using the R package clusterProfiler (v4.14.0) [31]. The results indicated that the candidate driver genes across ten cancer types were significantly enriched in between 128 and 191 related pathways. Notably, the top 15 enriched pathways for BLCA, BRCA, COAD, HNSC, and KIRC are shown in Fig. 5, while those for the other five cancer types are presented in Fig. S6. Detailed enrichment results for ten cancer types are provided in Tables S1 to S10.

Notably, PI3K-Akt signaling pathway (ID: hsa04151) significantly enriched across nine cancer types, where it regulates critical processes such as cell survival, proliferation, and metabolism. PI3K-Akt is implicated in the initiation, progression, and therapeutic resistance of various malignancies. The complexity of its network has been well

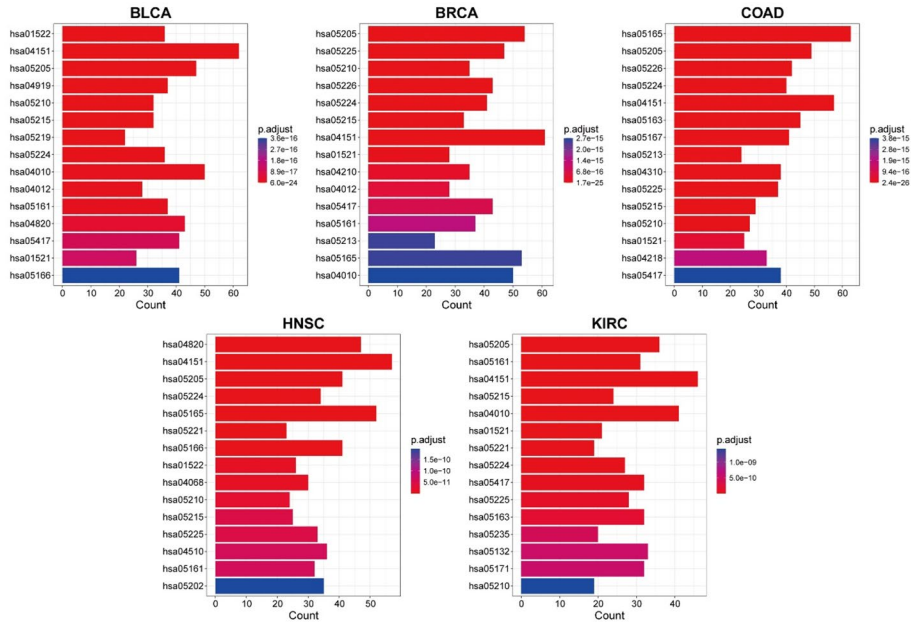

Fig. 5 KEGG cancer pathway enrichment analysis for candidate driver genes in BLCA, BRCA, COAD, HNSC and KIRC. The vertical axis indicates the KEGG cancer pathway ID

characterized, encompassing cascade activation mechanisms, upstream and downstream regulatory targets, and the development of specific inhibitors [32].

In BLCA, POLR3G activates PI3K-Akt signaling pathway ($p.adjust = 2.16 \times 10^{-22}$), promoting epithelial-mesenchymal transition (EMT), with its expression level significantly correlated with patient prognosis [33]. In UCEC, proteasome and PI3K-Akt signaling pathway ($p.adjust = 4.03 \times 10^{-15}$) are primarily involved in the cellular functions of RNF186, driving cancer proliferation, migration, and invasion [34]. Furthermore, in PRAD, PI3K-Akt signaling pathway ($p.adjust = 7.29 \times 10^{-20}$) has been shown to be molecularly associated with the development of castration resistance [35].

Proteoglycans in cancer (ID: hsa05205) also significantly enriched across nine cancer types, where it regulates critical processes such as tumor growth, metastasis, and angiogenesis. By interacting with ligands and receptors, proteoglycans modulate cell proliferation, survival, and blood vessel formation, thereby supporting tumor expansion. Its complex role extends to influencing the extracellular matrix, tumor-stromal interactions, and activating pathways that drive metastasis and therapeutic resistance [36].

In BRCA, Proteoglycans in cancer ($p.adjust = 1.66 \times 10^{-25}$) is primarily involved in the cellular functions of targets such as ALB, AKT1, and EGFR, which contributes to cancer cell proliferation and apoptosis [37]. In HNSC, EXT2 activates multiple oncogenic pathways, including Proteoglycans in cancer ($p.adjust = 8.88 \times 10^{-14}$), with elevated expression levels correlating with poor patient survival prognosis [38]. Furthermore, in COAD, Proteoglycans in cancer ($p.adjust = 3.26 \times 10^{-25}$) has been shown to be molecularly associated with cancer progression, influencing cell proliferation, migration, and invasion [39].

Subsequently, we identified CGC cancer driver genes within the candidate driver genes, which are likely to be further classified as tumor-specific. The CGC driver genes that ranked highly only in their corresponding cancer types are listed in Table S11. Notably, more than ten such top-ranked CGC driver genes were identified in HNSC,

KIRP, LUAD and UCEC. Figure 6 illustrates the potential tumor-specific driver interactions across various cancer types. The top-ranked CGC driver genes clearly contain rich tumor-specific information, offering valuable insights for further investigation into the intricate relationships between known driver genes and specific malignancies.

Prediction of tumor-specific driver genes

After excluding cancer driver genes from the CGC and IntOGen databases, we analyzed the top 10 remaining candidate driver genes for each cancer type. A comprehensive literature review using PubMed revealed that most of these candidate genes are strongly associated with their respective cancers. This association is primarily reflected by their involvement in key signaling pathways that regulate cancer initiation and progression, or their roles as significant biomarkers of therapeutic resistance. The corresponding PubMed identifiers (PMIDs) for the relevant studies are summarized in Table S12. Notably, genes such as GAPDH, ACTB, TTN, MAPK3, and MUC16 consistently ranked among the top predictors across various cancer types.

The top 10 remaining candidate driver genes, ranked highly in only a few cancer types, are considered potential tumor-specific driver genes. Notably, RYR2 exhibits significant mutation differences in the immune subtype of BLCA [40]. PI3K inhibition enhances anticancer effects via cytotoxic autophagy in BRCA [41]. LGR5 is a crucial marker involved in the metastasis and progression of COAD [42]. Mutations and dysregulation of APEX1 are linked to an increased risk of HNSC [43]. The tyrosine-metabolizing enzyme GSTZ1 is recognized as a biomarker for KIRC [44]. The silencing of ASPM attenuates the expression of TCF4 in KIRP [45]. ALB plays a pivotal role in the subtype-intrinsic drive of LUAD [46]. HMGB1 is positively correlated with the improved prognosis in LUSC [47]. MMP9 regulates NET formation through neutrophil activation in

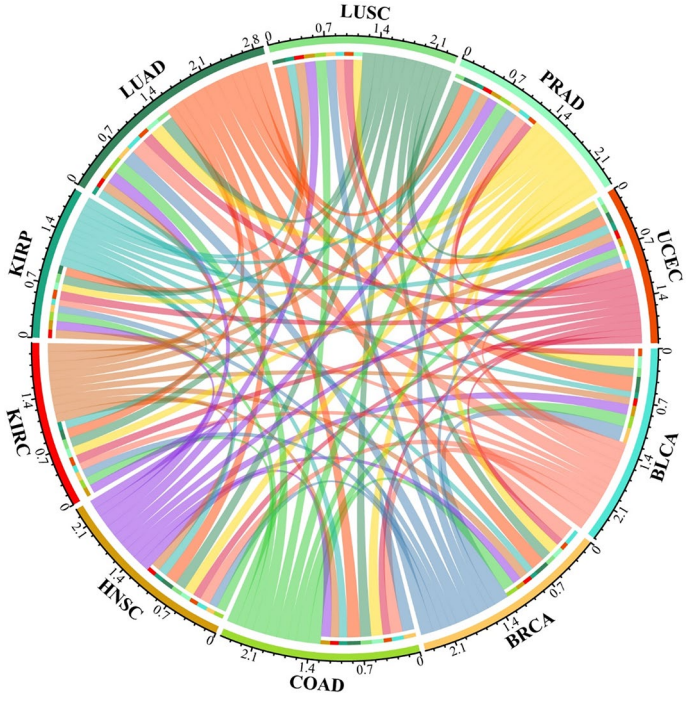

Fig. 6 Potential interactions of top-ranked CGC driver genes across various cancer types. These driver genes are likely to be further classified as tumor-specific

PRAD [48]. Furthermore, TNF can guide the selection of the immunotherapy for UCEC [49].

Analysis of tumor-specific driver genes

Using the online platform GEPIA2 [50], we conducted survival analysis for the potential tumor-specific driver genes. For example, EGF and PCLO in BLCA, DES, IL6, and ITGA5 in HNSC, LRP8 and THY1 in KIRP, KRT18 and PKP2 in LUAD, and CXCL12 in LUSC were selected. As shown in Fig. 7, significant survival differences were observed between the high- and low-expression groups for specific cancer types ($P < 0.05$, $HR > 1$), indicating a potential association between the expression of these driver genes and the progression of their respective cancers. The analysis results for the remaining tumor-specific driver genes are presented in Fig. S7.

Furthermore, in tumor samples from other cancer types, the majority of high- and low-expression groups did not show significant survival differences (see Fig. 8). Notably, KRT18 ($p(HR) = 0.042$) was the gene to exhibit significant survival differences only in LUAD tumor samples. This finding suggests that KRT18 exhibits strong specificity for LUAD, highlighting its potential as a novel biomarker and therapeutic target [51]. Consequently, analyzing these tumor-specific driver genes is crucial for the early diagnosis, targeted therapies, and precision medicine of specific cancers.

Additionally, we conducted differential expression analysis for the potential tumor-specific driver genes using the R package limma [52]. For example, PI3 in BRCA, DES in HNSC, MAOB in KIRC, PXDNL in KIRP, and CXCL12 in LUSC were selected. As shown in Fig. 9, significant differential expression of these driver genes was observed across subtypes within their respective cancers ($P < 0.05$, $|\log_2 FC| > 2$), suggesting that these genes play a pivotal role in revealing tumor heterogeneity across distinct subtypes. Therefore, analyzing these tumor-specific driver genes is essential for subtype stratification, prognostic evaluation, and personalized therapies of specific cancers.

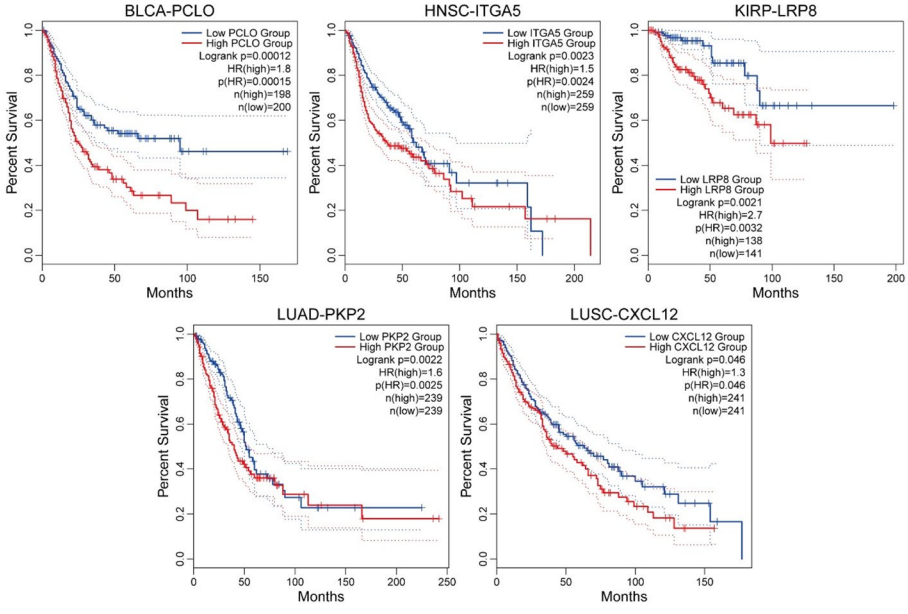

Fig. 7 Survival analysis of selected potential tumor-specific driver genes. The survival curve for high-expression group is depicted in red, while low-expression group is depicted in blue. Dashed lines indicate the confidence intervals for each respective group

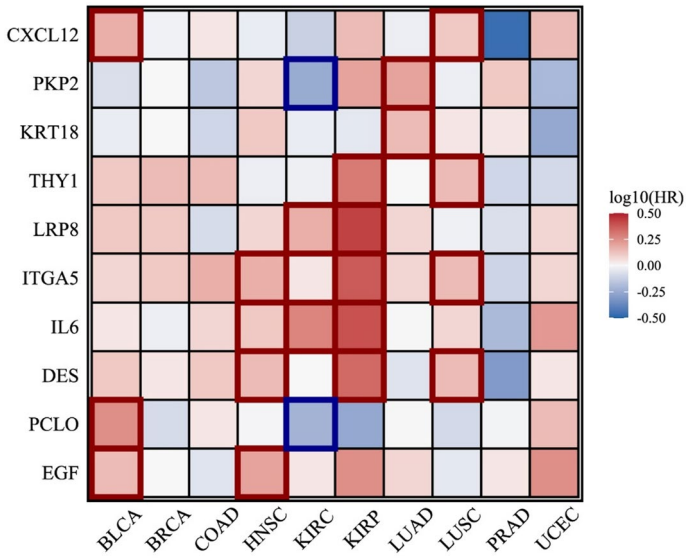

Fig. 8 HR heatmap for survival analysis of potential tumor-specific driver genes. Bold borders indicate significant survival differences in the corresponding tumor samples

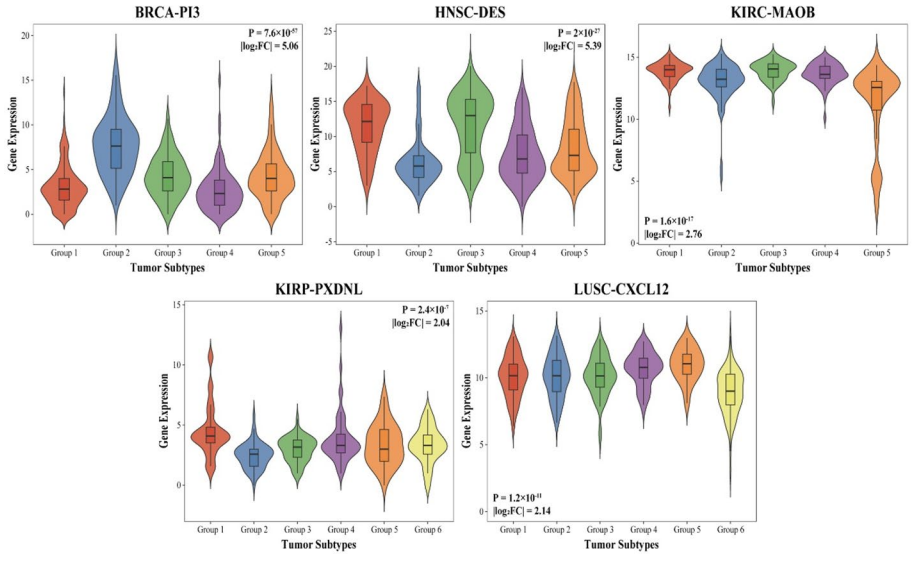

Fig. 9 Differential expression analysis of selected potential tumor-specific driver genes. The horizontal axis illustrates the subtype stratification results for their respective cancers

Discussion

In the pursuit of accurately identifying key driver genes for specific cancer types, a critical challenge in bioinformatics research, we present MODCAN, a novel semi-supervised algorithm based on multi-omics features and differential co-association networks. MODCAN comprehensively explores multi-omics features that reflect the inherent heterogeneity of tumors and constructs differential co-association networks based on distinct tumor subtypes, thereby facilitating the complementary integration of information.

When applied to ten cancer datasets from TCGA, MODCAN significantly outperforms both existing supervised and unsupervised learning algorithms, exhibiting superior performance in terms of precision, recall, and AUPR. Furthermore, MODCAN

demonstrates substantial advantages in predicting potential tumor-specific driver genes. Notably, these genes not only exhibit strong specificity for their respective cancers, but also reveal tumor heterogeneity across distinct subtypes. The innovative contributions of MODCAN are particularly evident in three key areas:

(i) MODCAN integrates gene expression, methylation data, and community cohesion scores to facilitate the meaningful stratification of tumor samples into distinct subtypes. These subtypes exhibit significantly different survival probabilities in Kaplan-Meier curves.
(ii) MODCAN extracts global features from gene features based on the complete genomic data and network features calculated from the PPI network, while concurrently deriving local features from the genomic data specific to the corresponding tumor subtypes.
(iii) To model shifts in gene correlations within tissue samples during carcinogenesis, MODCAN constructs differential co-association networks based on the PPI network by exploiting the Pearson correlation differences between tumor and normal samples.

MODCAN outperforms current advanced algorithms in identifying cancer driver genes; however, it still has certain limitations. First, in calculating network features based on the PPI network, MODCAN treats it as an undirected weighted network, overlooking the hierarchical relationships inherent in gene regulatory mechanisms. This simplification results in network features that fail to fully capture the complex gene interactions. Second, in constructing the differential co-association networks, MODCAN considers only the effects of expression and methylation data on gene interactions within tumor subtypes, without accounting for the influence of community cohesion scores. As a result, important subtype-specific information may be lost in the networks.

Conclusions

Future improvements to MODCAN will primarily focus on refining the differential co-association networks. One key objective is to integrate community cohesion scores into the network construction, enabling them to provide more precise guidance for capturing subtype-specific information. A parallel goal is to develop an explanatory framework that elucidates the pivotal roles of predicted driver genes within their respective networks, thereby enhancing both the interpretability and biological relevance of the results.

Despite these limitations, the superior performance of MODCAN in accurately identifying key driver genes for specific cancer types suggests that it represents a valuable computational framework with broad potential applications in cancer research, including early diagnosis, prognostic evaluation, and precision medicine.

Abbreviations

TCGA	The cancer genome atlas
COSMIC	Catalogue of somatic mutations in cancer
CGC	Cancer genome catalogue
NCG	Network of cancer genes
OMIM	Online mendelian inheritance in man
KEGG	Kyoto encyclopedia of genes and genomes
SNV	Single nucleotide variant
CNV	Copy number variation
PPI	Protein-protein interaction
EMT	Epithelial-mesenchymal transition
PMID	PubMed identifier

MRS	Median rank score
MGCN	Multi-graph convolutional network
BLCA	Bladder urothelial carcinoma
BRCA	Breast Invasive Carcinoma
COAD	Colon adenocarcinoma
HNSC	Head and neck squamous cell carcinoma
KIRC	Kidney renal clear cell carcinoma
KIRP	Kidney renal papillary cell carcinoma
LUAD	Lung adenocarcinoma
LUSC	Lung squamous cell carcinoma
PRAD	Prostate adenocarcinoma
UCEC	Uterine corpus endometrial carcinoma

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12859-025-06331-w.

Supplementary Material 1

Supplementary Material 2

Supplementary Material 3

Supplementary Material 4

Supplementary Material 5

Supplementary Material 6

Supplementary Material 7

Supplementary Material 8

Supplementary Material 9

Supplementary Material 10

Supplementary Material 11

Supplementary Material 12

Supplementary Material 13

Supplementary Material 14

Supplementary Material 15

Supplementary Material 16

Supplementary Material 17

Supplementary Material 18

Supplementary Material 19

Supplementary Material 20

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Author contributions

P.L. and G.X. are the joint authors. P.L. is responsible for the project's conception. P.L. and G.X. are the major contributors in writing the manuscript. P.L., G.X. and H.W. are the major contributors in analyzing the results. P.L. provided the programming support. H.W., C.X. and Y.Z. provided the helpful discussion for the project's progress. All authors have read and approved the final manuscript.

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Data availability

The datasets analyzed in this study for TCGA cancer patients, obtained from the UCSC Xena Browser (https://xenabrowser.net/datapages/), have been organized and are available in the OSF repository at https://doi.org/10.17605/OSF.IO/FWGM9. The dataset IDs from the UCSC Xena Browser for each cancer type are as follows: BLCA: TCGA-BLCA.star_counts.tsv, TCGA-BLCA.methylation450.tsv; BRCA: TCGA-BRCA.star_counts.tsv, TCGA-BRCA.methylation450.tsv; COAD: TCGA-COAD.star_counts.tsv, TCGA-COAD.methylation450.tsv; HNSC: TCGA-HNSC.star_counts.tsv, TCGA-HNSC.methylation450.tsv; KIRC: TCGA-KIRC.star_counts.tsv, TCGA-KIRC.methylation450.tsv; KIRP: TCGA-KIRP.star_counts.tsv, TCGA-KIRP.methylation450.tsv; LUAD: TCGA-LUAD.star_counts.tsv, TCGA-LUAD.methylation450.tsv; LUSC: TCGA-LUSC.star_counts.tsv, TCGA-LUSC.methylation450.tsv; PRAD: TCGA-PRAD.star_counts.tsv, TCGA-PRAD.methylation450.tsv; UCEC: TCGA-UCEC.

star_counts.tsv, TCGA-UCEC.methylation450.tsv. The source code to train the MODCAN model and reproduce the results is available at https://github.com/Kevin-Lpn/MODCAN.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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